

PHUBAI, Viet Nam

WMO: 488520

Lat: 16.402N

Lon: 107.703E

Elev: 48

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.6	59.3	55.3	65.2	60.6	57.1	69.7	61.2	14.5	71.4	13.5	71.8	4.0	270	0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	16.8	99.0	81.0	97.1	80.4	95.4	80.1	83.8	93.8	82.6	92.3	81.7	91.0	8.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.4	163.7	87.9	80.5	158.6	86.8	79.2	152.0	85.5	48.0	94.9	46.7	92.2	45.7	91.3	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	11.9	10.2	DB	55.0	102.3	4.0	3.2	52.1	104.6	49.7	106.5	47.5	108.2	44.5	110.6	(4)
(5)			WB	54.4	85.5	3.9	2.6	51.6	87.4	49.3	88.9	47.1	90.3	44.3	92.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	78.7	69.0	70.9	75.0	80.5	84.2	86.2	85.7	85.4	82.3	79.1	75.3	70.3	(6)
(7)		DBStd	7.43	4.85	6.06	5.94	4.99	3.72	2.75	2.60	2.81	3.08	3.00	4.20	4.82	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	54	20	18	5	0	0	0	0	0	0	0	1	10	(9)
(10)		CDD50	10470	588	586	774	914	1061	1085	1106	1097	970	902	759	628	(10)
(11)		CDD65	5048	143	183	314	464	596	635	641	632	520	437	310	173	(11)
(12)		CDH74	49646	467	921	2137	4424	6845	8177	8020	7658	5221	3410	1767	599	(12)
(13)		CDH80	21239	67	252	699	1823	3166	4172	3965	3693	1983	957	374	88	(13)
(14)	Wind	WSAvg	3.8	3.9	3.8	3.6	3.4	3.6	3.7	3.7	3.6	3.6	4.0	4.4	4.2	(14)
(15)	Precipitation	PrecAvg	122.4	5.0	2.2	2.4	2.6	4.8	3.3	3.3	6.8	17.1	31.6	25.7	16.6	(15)
(16)		PrecMax	214.2	13.7	5.7	7.0	8.0	12.7	9.1	9.1	24.2	51.3	58.6	87.4	45.3	(16)
(17)		PrecMin	81.8	1.3	0.2	0.4	0.3	1.1	0.6	0.8	1.2	2.2	12.8	6.4	3.5	(17)
(18)		PrecStd	31.3	3.5	1.5	2.0	2.1	2.8	2.1	2.3	5.2	10.6	12.2	17.0	10.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	91.4	94.4	98.6	100.6	102.2	100.8	101.7	97.1	93.4	89.7	86.1	(19)
(20)			MCWB	75.9	77.6	79.5	82.1	82.0	81.4	81.8	80.9	80.6	80.6	79.1	77.3	(20)
(21)		2%	DB	81.4	86.7	90.9	95.1	97.2	98.6	97.9	98.0	94.4	89.6	86.4	82.4	(21)
(22)			MCWB	73.5	77.1	79.2	81.3	81.4	81.2	80.0	80.1	80.1	79.5	78.2	75.4	(22)
(23)		5%	DB	78.7	82.6	87.4	92.6	95.2	96.5	95.5	95.3	91.7	87.3	84.2	79.3	(23)
(24)			MCWB	72.6	75.3	78.4	80.4	80.8	80.4	79.3	79.6	79.8	78.4	76.9	73.6	(24)
(25)		10%	DB	76.3	79.2	83.8	89.6	92.8	94.4	93.7	93.3	89.5	85.1	81.9	77.1	(25)
(26)			MCWB	71.5	73.6	76.9	79.6	80.3	80.1	79.3	79.3	79.5	77.7	76.0	73.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.8	79.4	81.4	83.9	84.5	85.0	85.4	85.2	82.6	82.2	80.1	78.3	(27)
(28)			MCDWB	83.1	87.8	90.7	95.1	94.4	95.3	95.4	95.1	91.8	88.7	86.1	84.0	(28)
(29)		2%	WB	75.0	77.4	79.8	82.0	82.8	83.1	82.8	82.6	81.5	80.4	79.0	76.4	(29)
(30)			MCDWB	79.5	84.5	88.1	92.1	93.4	93.1	91.7	91.7	89.8	86.8	84.2	80.4	(30)
(31)		5%	WB	73.6	76.0	78.7	80.9	81.7	81.9	81.5	81.3	80.7	79.3	78.1	75.0	(31)
(32)			MCDWB	77.2	81.6	85.7	89.9	91.8	91.7	90.4	90.1	88.3	84.7	82.6	78.2	(32)
(33)		10%	WB	72.2	74.5	77.4	79.9	80.8	81.1	80.5	80.4	79.9	78.6	77.1	73.6	(33)
(34)			MCDWB	75.3	78.5	82.8	87.7	90.2	90.5	89.7	89.0	86.8	83.3	80.8	76.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	11.4	12.4	14.2	15.8	16.8	16.4	16.3	13.1	11.3	9.6	8.5	(35)
(36)			MCDBR	15.1	17.7	18.4	19.4	19.8	20.1	19.6	19.8	17.6	15.7	14.0	13.5	(36)
(37)			MCWBR	7.7	8.4	8.5	8.1	7.1	6.9	6.6	6.4	6.3	6.4	6.3	6.6	(37)
(38)		5% WB	MCDWB	13.3	16.8	16.8	17.9	18.0	18.1	18.0	17.8	15.4	13.6	12.4	12.0	(38)
(39)			MCWBR	7.3	8.2	8.2	8.1	7.2	7.0	7.0	6.7	6.1	6.1	6.1	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.521	0.563	0.685	0.622	0.516	0.471	0.455	0.466	0.527	0.578	0.544	0.562	(40)	
(41)		taud	2.016	1.934	1.691	1.873	2.177	2.320	2.358	2.314	2.132	1.945	2.005	1.898	(41)	
(42)		Ebn at Noon	243	240	215	230	252	261	266	266	250	233	235	226	(42)	
(43)		Edh at Noon	53	60	79	66	48	41	40	42	50	59	53	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	939	1257	1378	1621	1768	1738	1634	1595	1432	1176	976	757	(44)	
(45)		RadStd	143	233	132	147	129	112	166	119	171	170	184	193	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	N/A	N/A	+1.69	+0.95	+0.90	N/A	N/A	+544	+536
(47) Regional (3 neighbors)		+0.68	N/A	N/A	+1.21	+0.54	N/A	N/A	N/A	+410	+407

Nomenclature: See separate page